

TabPFN for operational intrusion detection

- Prior approach** IDS benchmark data collected under daily attack scenarios
Train on all attack classes and aim to improve on baseline F1
- Deployment** New attack labels and changing conditions require model updates
F1 on a fixed test set alone cannot establish operational utility
- Why TabPFN** A pretrained tabular foundation model used directly for inference
Strong prediction on small data; context updates without retraining

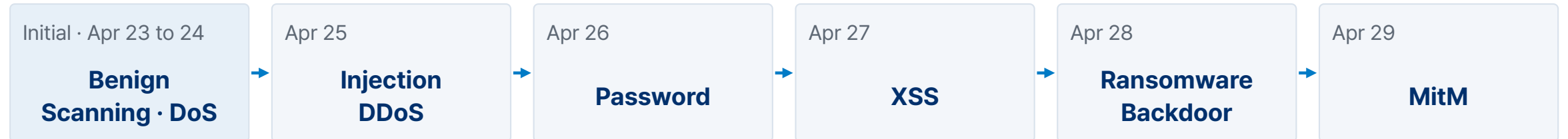
Updating an IDS as new attack labels arrive

XGBoost retraining vs. Global TabPFN context expansion · Same cumulative examples · Seed 43

CIC2018 · 2018



ToN · 2019



Final budget: 100,000 rows each · No future classes in training · All test rows of introduced classes

Original class arrival order · New attack performance, existing detection, update and inference cost

CIC2018 · New attack performance and cost

Same cumulative examples · Global TabPFN v3 · Seed 43 · RTX 4090 · GPU jobs run sequentially

F1 at first introduction (%)

Sorted by test count · Introduction stage in parentheses

New attack	Test count	XGBoost	TabPFN
DDoS (1)	262,680	99.96	99.53
Bot (4)	41,542	99.70	99.77
Infiltration (3)	8,742	10.68	25.43
Web attack (2)	127	17.32	31.52

Final stage: 100,000 training examples

Full test: 3,042,473 rows · Time in seconds

Metric	XGBoost	TabPFN
Macro-F1 (%)	77.59	78.64
Benign FPR (%)	2.10	0.14
Fit / context setup	1.98	16.98
Inference	1.44	522.63

TabPFN: fixed weights; preprocessing and cache setup
XGBoost: retrained on cumulative data at each stage

Higher initial F1 for Web attack and Infiltration · Reduce inference cost with large contexts

ToN · New attack performance and cost

Same cumulative examples · Global TabPFN v3 · Seed 43 · RTX 4090 · GPU jobs run sequentially

F1 at first introduction (%)

Sorted by test count · Introduction stage in parentheses

New attack	Test count	XGBoost	TabPFN
XSS (3)	343,854	87.23	82.62
Password (2)	210,951	88.86	88.99
DDoS (1)	103,338	96.88	96.98
Injection (1)	68,891	68.90	72.71
Backdoor (4)	40,030	99.87	99.91
MitM (5)	1,161	15.26	4.66
Ransomware (4)	709	14.52	12.47

Final stage: 100,000 training examples

Full test: 2,271,723 rows · Time in seconds

Metric	XGBoost	TabPFN
Macro-F1 (%)	71.30	69.51
Benign FPR (%)	11.03	13.33
Fit / context setup	3.59	17.15
Inference	1.61	391.50

TabPFN MitM · Recall 96.99% · Precision 2.39%
41,330 benign samples among 46,010 false positives

High MitM recall, many benign false positives · Improve context composition and inference cost

Data review: remove rows with conflicting labels

Identical features · Rows: original labels · Columns: other labels · Cells: row-class share (%) · Sorted by original class size

CIC2018

Conflicting rows 26.65%

	Ben	DDoS	Brute	DoS	Bot	Inf	Web
Benign		.		.		26.7	.
DDoS	.						
Brute force				67.2			
DoS	.		34.8				
Bot							
Infiltration	75.1						.
Web attack	69.5					17.0	

ToN-IoT

Conflicting rows 60.73%

	Ben	DDoS	XSS	Pass	Scan	Inj	DoS	Back	MitM	Ran
Benign		13.7	3.3	.	49.9	.	8.6	2.3	8.2	2.4
DDoS	76.3		69.1	22.4	1.4	75.0	.	.	.	.
XSS	37.0	12.2		15.4	.	29.4	6.8	.	.	.
Password	33.3	.	.		.	.	.	.	.	.
Scanning	97.1	.	.	.		.	.	.	.	3.6
Injection	7.5	3.4	9.0	2.1	2.0		2.0	.	.	.
DoS	33.1	12.8	26.3	.	9.1	12.2		.	1.6	
Backdoor	1.3	.	.	.	.	.	.			.
MitM	8.5	3.1	7.2	2.1	2.8	5.3	7.5			
Ransomware	9.2	.	1.0		1.9	.		.		

BoT-IoT

Conflicting rows 6.11%

	DoS	DDoS	Recon	Ben	Theft
DoS		2.0	1.2	1.2	
DDoS	1.6		.	.	
Recon	2.5	.		35.6	1.4
Benign	7.7	.	63.8		
Theft			67.6		

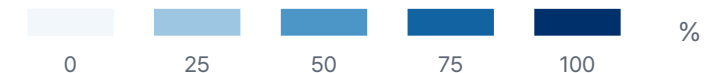
UNSW-NB15

Conflicting rows 1.33%

	Ben	Expl	Fuzz	Gen	Recon	DoS	Back	Shell	Anal	Worm
Benign										
Exploits			6.0	4.6	4.7	3.9	4.0	3.6		.
Fuzzers		18.0		15.2	15.8	13.4	14.0	13.0		.
Generic		4.6	1.4		.	.	66.6	.		.
Recon		30.5	31.0	3.9		29.4	29.6	3.4		
DoS		9.2	9.1	8.4	8.6		8.6	7.5		
Backdoor		18.1	18.7	19.1	17.1	16.7		13.7		.
Shellcode		5.8	6.6	3.5	3.7	3.3	11.8			
Analysis										
Worms		.	.	4.4			4.4			

Blank: 0% · Dot: <1% · Grey: same class

Overlapping counts; cell percentages cannot be summed



Web attack: 1,764 + 432 – 429 overlapping rows = 1,767 removed · 771 retained

Class distribution after cleaning and the fixed test set

Before/after: full dataset · Test: fixed cleaned holdout · Sorted by class count after cleaning

CIC2018

Class	Before cleaning	After cleaning	Test
Benign	17,514,626	12,798,328	2,652,203
DDoS	1,324,350	1,316,025	262,680
Bot	207,703	207,703	41,542
DoS	302,966	197,415	39,483
Brute force	575,194	188,414	37,696
Infiltration	188,152	46,761	8,742
Web attack	2,538	771	127
Total	20,115,529	14,755,417	3,042,473

ToN

Class	Before cleaning	After cleaning	Test
Benign	16,792,214	6,446,120	1,474,302
XSS	2,834,435	1,670,497	343,854
Password	1,594,777	1,061,699	210,951
DDoS	4,141,256	925,247	103,338
Injection	381,777	341,263	68,891
Backdoor	203,384	200,688	40,030
DoS	203,456	113,655	22,458
Scanning	1,358,977	39,069	6,029
MitM	6,013	5,446	1,161
Ransomware	3,971	3,604	709
Total	27,520,260	10,807,288	2,271,723

Global and expert models built from contexts

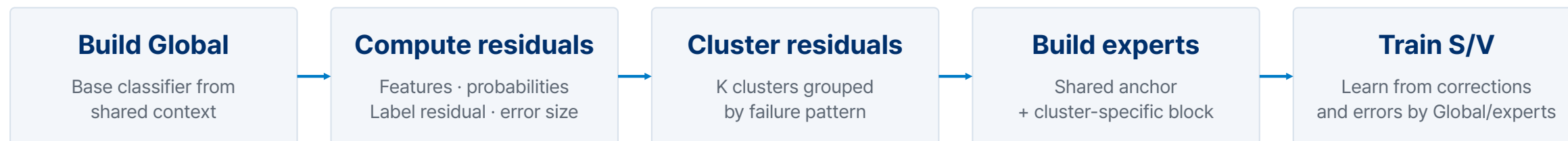
Shared and specialized contexts · Frozen TabPFN backbone · Scorer and verifier trained separately

Global pool · shared context

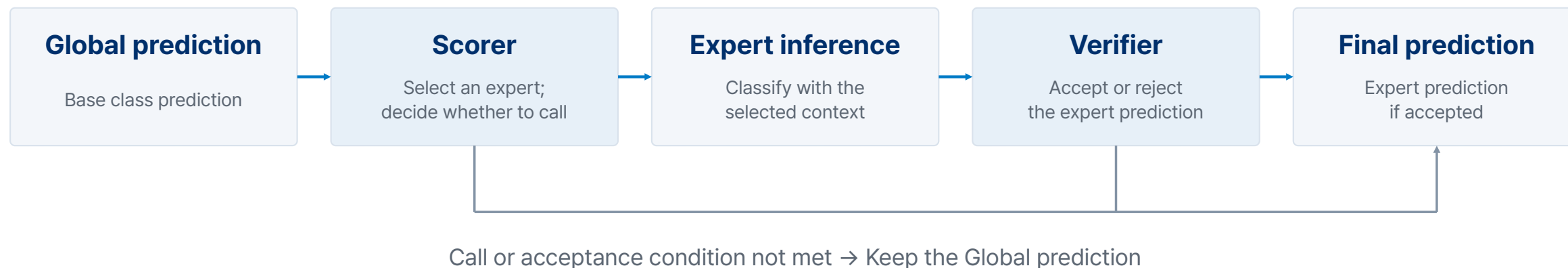
Expert pool · residuals and experts

Route pool · S/V training

Offline · model construction



Online · per-input prediction



Current static performance · SOTA comparison

Static setting with all attack classes in training · Same cleaned test set · Seed 43 · Macro-F1 (%)

Method	CIC2018	ToN	Training data
Global TabPFN v3	78.24	67.96	Shared 100,000 rows
XGBoost	78.21	69.83	Shared 100,000 rows
BoostPFN	71.48	58.66	Shared 100,000 rows
LoCalPFN · FT	70.24	64.62	Shared 100,000 rows
DistPFN · v3	78.39	66.82	Shared 100,000 rows
XGBoost · full train	77.95	71.62	Full training set
Proposed model · K = 4 · S/V	78.24	76.17	Global + Expert and Route pools

Proposed model: existing anchor + residual blocks · Additional labels from Expert and Route pools

Current static performance · Separate from the Global comparison under sequential attack introduction

SOTA cost on cleaned data

RTX 4090 24GB · Up to 2 concurrent jobs · Elapsed time includes resource contention

Method	CIC fit/setup s	CIC inference s	ToN fit/setup s	ToN inference s	CIC GPU GiB	ToN GPU GiB
Global TabPFN v3	36.4	1,099.3	36.7	821.7	2.92	2.94
XGBoost	2.3	1.0	55.4	1.0	0.55	0.53
BoostPFN	98.5	2,308.5	97.8	1,737.6	0.89	0.89
LoCalPFN · FT	3,196.3	17,300.8	5,783.1	11,144.8	7.54	7.54
DistPFN · v3	36.4	1,099.4	36.7	821.9	2.92	2.94
XGBoost · full train	107.8	1.0	143.2	1.1	2.20	1.59

LoCalPFN: validation in fit, retrieval in inference · DistPFN: Global computation plus prior correction

Current K = 4 model · Policy call rate: CIC2018 0.00%, ToN 100.00% · Conditional runtime to be measured

Next steps · An IDS that incorporates new attacks

Gains on some new CIC2018 attacks · Reduce ToN false alarms and computational cost on both datasets

New attack adaptation

Same newly labeled examples
Per-attack performance after updates
Changes in rare-class detection

Preserve existing detection

Same existing test samples
Corrections, errors and benign FPR
Effect of context composition

Operational cost

Update cost when examples arrive
Inference cost until the next update
Throughput and memory

**Research criterion: how well new attacks are incorporated,
and at what cost existing detection performance is preserved**

Vary benign examples and context size → Compare false alarms and cost → Test expert/S/V updates

F1 is one measure of system utility · Unknown-attack recognition is a later OOD extension